Lo Barnechea, Santiago

Here El Niño arrives as rain on the mountain, not heat: debris flows out of the quebradas now, fire on the foothill edge in summer. The first move is a local evacuation trigger that does not wait for the state's cell alert.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Not resolved (contains the comuna, ~132,000 people)	No figure published for Chile	n/a (study covers Jun 2026 – Feb 2027)	Debris flow and flood (Oct–Dec); wildfire (Jan–Mar)

9/9 recommendations checked. 9 corrected, 0 unverified, 5 contraindicated. Every action survived, but every one was narrowed: no rain threshold for debris flows has been published, the borrowed effect sizes did not transfer, and two proposals duplicated things the water utilities already do. One item was downgraded from "do not" to a caution: the heat warning. Exposure figure: Climate Impact Lab, *451,000 at Risk*, September 2026, pre-peer-review. It publishes no figure for Chile or any containing geography (Brazil is the only South American country named), so no death count is given here and none has been scaled from another country. The figure covers heat only, and heat is not this place's main El Niño hazard. **The funding target fell by 41% in verification**, from USD 110,000 to USD 65,000. Fire detectors were dropped as a wildfire measure, refilling the drained fire-water pools turned out to be a municipal duty, and the Cruz Roja placement moved above the target because no local presence could be confirmed.

EXECUTIVE SUMMARY — AT A GLANCE



33.35°S 70.52°W, Chile · Natural Earth

- **The risk.** Warm storms send debris flows down the quebradas: 3 died at Ñilhue in 2009, and a flow next door in Las Condes destroyed 56 homes on 8 October 2026.
- **First action.** The municipality and SENAPRED RM adopt a field-sign and spotter evacuation trigger for quebrada homes by 31 October.
- **The money.** Floor USD 40,000, target USD 65,000, ceiling USD 125,000. Most lives per dollar: warning kit for the juntas de vecinos along the quebradas (USD 30,000).
- **Do not.** Do not wait for the SAE phone alert to leave a quebrada: none was sent before the 8 October flow.*

What to do	**9** actions	p. 2
Who does what	RACI chart	p. 10
Where money does the most	funding	p. 10
Do not do this	**5** items	p. 12
What has happened here	the record	p. 14
How this was produced	methodology	p. 16
References	sources	p. 18

* Suggested formulation: check with a health worker before sharing.

WHAT THIS PLACE SHOULD DO RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached independently is the stronger bet. Scope: the request named a street in La Dehesa; this briefing covers the whole comuna of Lo Barnechea, the smallest unit with data, because the quebradas, the water plant, and the fire edge that decide the risk on that street are comuna-wide.

1. Adopt a local evacuation trigger for quebrada homes that does not wait for the state's phone alert

PARTIALLY CONFIRMED | EMERGENCY MANAGER · FRONTLINE · SKEPTIC · HISTORIAN · PUBLIC HEALTH · PHILANTHROPIST

ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Municipalidad de Lo Barnechea, with SENAPRED RM	31 October 2026	Staff time, whistles, megaphones, paint	Deaths in homes on quebrada channels and fans

For homes on the channels and fans of the Ñilhue, Huallalolén, El Arrayán, Las Hualtatas, and Las Zorras quebradas, and along Ruta G-21 (Pastor Fernández at km 6, El Cajón, Paso Marchant): two or three resident spotters per quebrada on one WhatsApp group; a whistle or pot-banging signal; door-to-door evacuation along painted uphill routes, out of the channel line; and a written agreement that shelters take pets and livestock, because rural families on Camino La Disputada refused shelters in 2021 over their animals. The trigger is field signs (the estero runs brown, rumbles, or suddenly goes quiet) and the DGA river gauges the municipality already watches. Any rain-intensity figure, such as 10 mm/h, is a placeholder until Sernageomin validates one.

* Suggested formulation: check with a health worker before sharing.

Why it ranks first: the evidence is local. No SAE alert was sent before the 8 October 2026 flow in San Carlos de Apoquindo; SENAPRED said the event was sudden and that Chile has no monitoring able to anticipate a quebrada activating. In 2009 the Ñilhue flow killed three people. Corrected in verification: no published rain threshold exists for the Santiago foothills, and the borrowed effect sizes (Hallegatte's 4–36 benefit-cost ratio, which is for national weather services; Dowling and Santi's developing-country death tolls, when Chile falls in the developed group) do not apply here and are not used.

2. Put quebrada ownership in writing, and check and empty the debris nets before every storm

PARTIALLY CONFIRMED · EMERGENCY MANAGER · HISTORIAN · SKEPTIC · ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Municipalidad, with DOH-MOP	This week; then before each front	Existing crews and plant; repairs as found	A full net or blocked culvert sending a flow into homes

Quebrada by quebrada, a signed note naming who clears each inlet and culvert and who orders evacuation. Inspect the nine flexible steel ring nets the comuna installed after the January 2021 debris flow (about CLP 600 million, including Huallalolén), repair or replace damaged units, and empty them before every frontal system. DOH-MOP publishes the fill level of each sediment basin it runs.

Evidence: in full-scale field tests at Illgraben, Switzerland, ring nets held debris flows, but a full net becomes a small dam that the next flow overtops (one 2006 test retained about 1,000 of 15,000 m³). The ownership point is local: after the 8 October flow the Las Condes mayor said the quebradas were not the municipality's responsibility, and the five MOP barriers promised there on a 90-day build had not been installed. Corrected: no evidence was found that any Lo Barnechea net is at the end of its life, so this reads "repair damaged units", not "replace old ones".

3. Write down the rule for closing Ruta G-21 on warm rain, enforce it, and stock a shelter in Farellones

PARTIALLY CONFIRMED · EMERGENCY MANAGER · HISTORIAN · FOOD SUPPLY · ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Carabineros and MOP Vialidad; Municipalidad for the shelter	1 November 2026 (ski season runs into November)	Not costed	Cars stranded or caught on the road when a slide hits

* Suggested formulation: check with a health worker before sharing.

Turn the closures Carabineros already make case by case into a written protocol: stop upward traffic whenever SENAPRED or the DMC forecasts rain with a high freezing level (SENAPRED already links a 0 °C line at 2,700–3,600 m to slide risk). Enforce it at the gate, and pre-stock a shelter in Farellones with water, food, and blankets for 72 hours.

Local evidence: on 8 October 2026 the road was cut in four places, and the mayor denounced dozens of drivers pushing to go up despite the closure. In 2009 visitors were held overnight at the ski centres (about 200 vehicles by one account, 40–50 people by contemporary press). Corrected: closures already happen, so the gap is the written trigger and enforcement, not the idea.

4. Agree water collection points with all three utilities, and tell people to store water on the rain alert, not the cut notice

PARTIALLY CONFIRMED · EMERGENCY MANAGER · ECONOMIST · HISTORIAN · FOOD SUPPLY · FRONTLINE · PUBLIC HEALTH · ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Municipalidad, with Aguas Andinas, Aguas Cordillera, and Aguas Manquehue	15 November 2026	Utilities' own contingency budgets	Days without safe water for cooking, hygiene, and medical needs

Ask each utility for its comuna-specific list of tanker and tank points (puntos de abastecimiento), and publish it before the next front. Pre-register households that cannot queue or carry water, using the existing electro-dependent lists as a data source. When SENAPRED announces heavy rain, households fill clean, covered, dated containers; by the time a turbidity notice goes out, it is late.

* Suggested formulation: check with a health worker before sharing.

Local evidence: when the rivers run thick with mud the plants stop. In February 2021 about 4,000 El Huinganal customers were cut for up to 24 hours because the San Enrique plant could not treat Mapocho water at 7,800 NTU; in April 2016 Aguas Cordillera cut 19,870 customers in La Dehesa. The Pirque reserve tanks protect the Maipo system; how far they protect the Mapocho-fed zones could not be confirmed. Corrected: the comuna has three utilities, not one; the tanker playbook and an emergency plan already exist, so asking SISS to "require a plan" was dropped; and Chile's electro-dependency law (21.304) covers electricity only, not water.

5. Cut grass firebreaks on the Cerro 18 and Chaguay edge before the spring grass dries, refill the drained fire-water pools, and map walking exits

PARTIALLY CONFIRMED | EMERGENCY MANAGER · HISTORIAN · SKEPTIC · FRONTLINE · PUBLIC HEALTH · PHILANTHROPIST

ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Municipalidad, with CONAF, Bomberos (CBS), Carabineros, and juntas de vecinos	15 December 2026; patrols January–March	Not costed	Fire reaching single-exit hillside streets, and fire engines that cannot get up

Cut and maintain grass breaks at the Cerro 18 and Pie Andino/Chaguay interface, with junta "mingas" clearing pasaje edges. Refill the two forest-fire water pools on Cerro 18 that were drained during the December 2025 eviction, once their owner is confirmed. Map a walking exit for every pasaje, and patrol access roads for arson from January to March, as the municipality did after the 2024 fire.

* Suggested formulation: check with a health worker before sharing.

Evidence: a wet year followed by a dry summer drives fire activity in central Chile (Urrutia-Jalabert et al. 2018, Ecosphere), and this spring is the wettest on record so far. In January 2024 a fire of about 900 ha burned from Cerro Chaguay into Colina; containers of accelerant were found and arson is presumed. Corrected: firebreaks mainly help by giving firefighters access and are weak evidence against home losses (Syphard et al. 2011), and no source names who owns the drained pools, so who pays to refill them is unconfirmed.

6. Build a register of people who cannot leave on their own, each with a named neighbour, tied to the summer heat alerts

PARTIALLY CONFIRMED | FRONTLINE · PUBLIC HEALTH · EMERGENCY MANAGER | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Municipalidad (DIDECO, Protección Civil), with juntas de vecinos	30 November 2026	Staff and volunteer time	Older and disabled people left behind in an evacuation or alone in a heat wave

List older people living alone (in La Dehesa, reached through building concierges and domestic workers), people with limited mobility, households without a smartphone or with a foreign SIM, and rural families on Corral Quemado and Camino La Disputada. Give each a named neighbour who knocks during an evacuation. From November to March, a red heat alert under the national heat annex (Senapred and MINSAL) triggers daily phone checks on people over 60 who live alone. The comuna's 2026 older-adults programme is the ready channel.

* Suggested formulation: check with a health worker before sharing.

Evidence: the 2017 and 2019 heat waves caused 584 and 245 excess deaths across Chile, mostly among over-65s and in the Metropolitan Region (UC / Lancet Countdown brief). About 14% of the comuna is 60 or over. Corrected: Ahmedabad's heat plan (about 1,190 deaths avoided a year) is a proof of concept from a far hotter, poorer city of 6 million, not an effect size for here. No Lo Barnechea register of vulnerable people was found, and no national phone-check programme exists, so this is a real addition.

7. Find and follow the families cleared from Quebrada Las Zorras, and put them on the alert lists

PARTIALLY CONFIRMED | SKEPTIC · PHILANTHROPIST · FRONTLINE | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Municipalidad (social follow-up), with Techo-Chile	Contact by 30 November; review at 6 and 12 months	Not costed	Displaced families resettling in another quebrada or fire edge, unseen

Quebrada Las Zorras on Cerro 18 once held 132 homes and 326 people. Families left in stages: 29 voluntarily in 2024, 36 homes taken down by December 2025, and the last 30 homes in early October 2026. The municipality pays rent subsidy for up to 24 months with social follow-up; Techo-Chile, which took part in the December 2025 operation, adds tracking at 6 and 12 months, so it is known whether each family is now safer, and puts them on the spotter and register lists in actions 1 and 6.

* Suggested formulation: check with a health worker before sharing.

Why it matters: the eviction was justified by fire, flood, and landslide risk, but no public reporting says where the families went, and Techo-Chile has said the CLP 340,000–350,000 monthly subsidy does not rent anything legal in the comuna. Corrected: networked heat detectors (the Lumkani model from South African settlements) were proposed here, but their evidence is self-reported and for house-to-house shack fires, they are not sold in Chile, and they give no useful warning of a wind-driven grass fire. They are dropped from this action.

8. Have the Concejo create a ring-fenced emergency budget line with written release conditions, so money for evacuation moves on a forecast, not on a damage survey

PARTIALLY CONFIRMED | ECONOMIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Concejo Municipal and the Alcalde, with DIDECO	**30 November 2026**	**Illustrative CLP 500 million (not costed; about 0.3% of the 2024 budget)**	**Weeks of delay in paying for transport, clearing, and temporary rent**

The Concejo approves a budget modification creating an emergency line, and an agreement plus a DIDECO social-aid regulation sets when it is spent: a SENAPRED red alert together with a red threshold published by the DGA at Río Molina antes junta San Francisco (the gauge behind the 6 October yellow alert) or Río Mapocho en Los Almendros. It pays for evacuation transport, quebrada clearing, and temporary rent. DIDECO checks in advance that campamento and quebrada households are on the Registro Social de Hogares and have a CuentaRUT account, and trains staff to fill in the damage form (FIBE) quickly. Separately, the comuna drafts donation agreements under Ley 20.444 with Anglo American (Los Bronces is inside the comuna) and the ski resorts, naming the works in advance.

Why: nothing that protects Lo Barnechea from a debris flow is pre-arranged. Chile's World Bank catastrophe bond covered earthquake and tsunami only (and matured in March 2026); Cruz Roja Chilena's wildfire early-action protocol triggers in Araucanía, 600 km south; the regulator only authorised parametric cover in August 2025, and no product for a comuna was found. The state's Bono de Recuperación (CLP 375,000 to 1,500,000 per household, in four tiers) pays only after a FIBE survey. The municipality made four budget modifications in

* Suggested formulation: check with a health worker before sharing.

2024, so the mechanism is routine. Corrected: a trigger is a policy condition, not self-executing (spending still needs a mayoral decree); FIBE cannot be filled in before an event; and Ley 20.444 donations only apply once a catastrophe zone is declared, which on 8 October covered Las Condes only.

9. Keep school meals running through the summer holidays in Bajo Lo Barnechea and Cerro 18, and pre-stock food for evacuated families

PARTIALLY CONFIRMED | FOOD SUPPLY | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Schools, with the Municipalidad (DIDECO and emergency office)	December 2026	Within JUNAEB's budget; food boxes not costed	Children losing their daily meal in the fire months; a food gap in the first 72 hours after an evacuation

JUNAEB's school meal programme (PAE) already covers the holidays, and schools, not families, manage it, so ask the schools to keep it running. DIDECO can also bid to run JUNAEB's Escuelas Abiertas de Verano, though it is small (about 6,000 students nationally in 2026). The emergency office pre-stocks food boxes for families who have to leave home.

Honest signal: the El Niño food signal here is weak. Santiago does not run out of food; in June 2023 floods pushed wholesale lettuce up 32% and potato 27% in a week, but the shock passed quickly. The real break is the last 200 metres: a family with a full pantry, no tap water, no gas, and a dead fridge cannot eat, which is why water (action 4) ranks above food. Corrected: "Vacaciones en mi Escuela" does not exist; the box count and cost are unsourced.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Municipalidad	SENAPRED RM	MOP (DOH, Vialidad)	Carabineros	Water utilities	CONAF	Bomberos (CBS)	Juntas de vecinos	Techo-Chile	Schools
1. Quebrada evacuation trigger	**A/R**	**C**	**I**	**I**			**I**	**R**	**I**	
2. Quebrada ownership and nets	**A**	**C**	**R**					**I**		
3. Ruta G-21 closure rule	**R**	**C**	**R**	**A/R**						
4. Water points and storage	**A**	**C**			**R**			**R**		
5. Firebreaks, pools, exits	**A**	**C**		**R**		**R**	**R**	**R**		
6. Register of people who cannot leave	**A/R**	**C**						**R**		
7. Follow cleared families	**A**							**C**	**R**	
8. Ring-fenced emergency budget line	**A/R**	**C**								
9. Summer meals and food boxes	**A**									**R**

What the chart shows: the Municipalidad is accountable for eight of nine rows, and that is the point. It holds the money (a CLP 165 billion budget in 2024) and the authority, and SENAPRED's own explanation of 8 October was that the state cannot see a quebrada activating in time. Only the road rule sits with Carabineros. Rows 1, 2, 3, and 7 need a decision and a signature more than money.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR

USD 40,000 (CLP 38.1 million)

TARGET

USD 65,000 (CLP 61.9 million)

CEILING

USD 125,000 (CLP 119.0 million)

LARGEST UNCERTAINTY

Where the cleared Las Zorras families now live

Arithmetic, at CLP 952.00 per USD (Banco Central dólar observado, 22 September 2026; check before use). Floor = quebrada warning kit 30,000 + family tracking 10,000 = USD 40,000. Target = floor + Bomberos wildland kit 25,000 = USD 65,000. Ceiling = warning kit at 40,000 + Bomberos at 50,000 + tracking at 20,000 + Cruz Roja water storage 15,000 (only if a local presence is confirmed) = USD 125,000; above that, the money buys infrastructure the municipality can and should pay for. Every amount is an estimate built from what it buys, not a published unit cost: no cost-per-life figure exists for community debris-flow warning in a setting like this, and the World Bank figure first offered (Hallegatte 2012) is for national weather services in developing countries. **The target fell by 41% in verification**, from USD 110,000 to USD 65,000.

* Suggested formulation: check with a health worker before sharing.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Juntas de vecinos along the Ñilhue, Huallalolén, El Arrayán, and Las Hualtatas quebradas	Action 1: whistles, megaphones, VHF radios, route paint, spotter training	USD 30,000 (CLP 28.6M)	Direct grant	Estimate of equipment for five quebradas. Local case: no SAE alert before the 8 Oct 2026 flow	PARTIALLY CONFIRMED
Techo-Chile	Action 7: locate and follow up the roughly 95 cleared Las Zorras households at 6 and 12 months; enrol them on alert lists	USD 10,000 (CLP 9.5M)	Direct grant	Estimate: one field worker for about four months. Techo is active on Cerro 18 and took part in the December 2025 operation	PARTIALLY CONFIRMED
Cuerpo de Bomberos de Santiago (company covering Lo Barnechea to be confirmed)	Action 5: wildland firefighting kit for the Cerro 18 and Chaguay edge	USD 25,000 (CLP 23.8M)	Direct grant (tax route under the donations law unconfirmed)	Estimate. In January 2022, 29 engines were sent to Cerro 18 and narrow streets kept them back	PARTIALLY CONFIRMED
Cruz Roja Chilena (ceiling only)	Action 4: water containers for isolated older people during turbidity cuts	USD 15,000 (CLP 14.3M)	Pre-positioned stock	Estimate. Averts harm more than deaths; no local branch found	UNVERIFIED

Dropped in verification: networked fire detectors for 300 households (USD 25,000; see "Do not do this"), and refilling the two drained 40,000-litre fire-water pools on Cerro 18, which are municipal and cheap to refill. The funder's one thing: in a comuna with a CLP 165 billion budget, a philanthropic peso adds little as volume. Its value is in reaching people public money will not protect, because the municipality is evicting them. Fund the fire edge and the displaced, not the river walls.

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here.

Do not wait for the SAE phone alert, or a regional red alert, before leaving a quebrada

CONTRAINDICATED

No SAE alert was sent before the debris flow at San Carlos de Apoquindo on 8 October 2026. SENAPRED's director called the event sudden and said Chile has no monitoring able to anticipate a quebrada activating; the minister said an alert without defined evacuation zones would cause confusion. In the February 2024 Viña del Mar fire, residents said the SAE messages arrived with the fire already on them, and about 300 phone antennas burned. The alert is worth having; it is not a trigger to wait for.

Instead: leave on field signs (brown, rumbling, or suddenly silent water; heavy rain with a high snowline) and on the spotter network in action 1, uphill and sideways out of the channel.*

Do not assume the nets, check dams, or sediment basins above you will hold

CONTRAINDICATED

A full net becomes a small dam that the next flow overtops; in one field test a net held about 1,000 of 15,000 m³. In 2022 the Santiago governor found the Quebrada de Macul basins holding material that should not have been there. On 8 October 2026 the barriers promised for Las Condes had not been installed. Nobody downstream is told when a basin is full.

* Suggested formulation: check with a health worker before sharing.

Instead: inspect and empty before every front, publish fill levels, and plan evacuation as if the works were not there (action 2).

Do not stay to defend the house, or try to drive out down a single-exit hillside street

CONTRAINDICATED

The February 2024 fire in Viña del Mar and Valparaíso killed 138 people (Fiscalía, 2025). Reporting and the IFRC operation report describe failed evacuation, gridlock, and rows of burned cars on hillside streets. No source was found that counts deaths among people who stayed or who died in cars, so this rests on reported patterns, not a measured toll. Cerro 18 has the same layout: in the January 2022 fire, 29 engines were sent and narrow streets kept them back.

Instead: leave on foot, early, at the first smoke, by the walking exit mapped in action 5, to a meeting point agreed in advance.*

Do not count an eviction from a risk zone as risk reduction until you know where the families went

CONTRAINDICATED

Quebrada Las Zorras was cleared on grounds of fire, flood, and landslide risk, backed by a fire-service inspection and a 2025 decree. No public record says where its roughly 95 households now live. Techo-Chile says the subsidy does not rent anything legal in the comuna, which makes another informal site, possibly another quebrada, a plausible destination. An eviction that moves a family from one hazard to another has reduced nothing.

Instead: track each household at 6 and 12 months and put them on the alert lists (action 7).

Do not buy home heat or smoke detectors as protection against wildfire

CONTRAINDICATED

Networked heat detectors from South African informal settlements were proposed for Cerro 18's fire edge. Their evidence is self-reported (a GSMA survey of 255 households in Cape Town) and concerns fires spreading from shack to shack; a heat-rise sensor indoors gives no useful warning of a wind-driven grass fire coming up a hillside, and the product is not sold in Chile. Money spent here is money not spent on firebreaks and exits.

* Suggested formulation: check with a health worker before sharing.

Instead: fund the firebreaks, refilled pools, and walking exits in action 5. Detectors may still be worth piloting for house fires, which Cerro 18 has had (2021, 2022), but not as a wildfire measure.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
June 1982 (El Niño)	Storm; the Mapocho overflows across Santiago, including Lo Barnechea, where about 30 people were evacuated	9–15 dead across Santiago (sources differ)	Not documented for the comuna
3 May 1993	Quebrada de Macul debris flow, eastern Santiago (Peñalolén, La Florida)	26 dead, 8 missing	MOP built sediment basins; in 2022 they were found holding material
6–7 Sep 2009 (El Niño)	Debris flow in Quebrada Ñilhue at km 6 of Ruta G-21, and a second flow at Paso Marchant	3 dead (a mother and her one-year-old daughter, and a mine guard); 24 injured	The Supreme Court later found that unauthorised earthworks in the quebrada made it worse
April 2016 (El Niño)	Turbidity shuts treatment plants; the Mapocho overflows in Providencia	No deaths; about 800,000 customers cut in 27 comunas, plus 19,870 in La Dehesa	Aguas Andinas built reserve tanks at Pirque (Maipo system)
January 2021	Debris flows at Yerba Loca (km 15, G-21) and Camino La Disputada	No deaths; at least 5 houses swept away, Corral Quemado cut off	Nine flexible ring nets installed in the comuna's quebradas (about CLP 600 million)
February 2021	Mapocho turbidity stops the San Enrique plant	About 4,000 El Huinganal customers without water for up to 24 hours	Nothing documented for the Mapocho-fed zones
January 2024 (El Niño)	Fire from Cerro Chaguay into Colina; accelerant found, arson presumed	About 900 ha; no homes lost	Municipal patrols of access roads

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
February 2024 (El Niño)	Viña del Mar and Valparaíso fire, 120 km away: the reference case for the foothill edge	138 dead	Prosecution of former CONAF officials; questions over SAE alert timing
8 Oct 2026 (El Niño)	Debris flow at San Carlos de Apoquindo, Las Condes, next door; Ñilhue and Huallalolén active in Lo Barnechea; G-21 cut in four places	No deaths; 6 injured, 384 left homeless, 56 homes destroyed, about 1,000 evacuated	Catastrophe zone declared for Las Condes only; red alert for slides covers Lo Barnechea

The recurrence nobody has noticed: the storms that kill here come out of season. September 2009, January 2021, and October 2026 were all warm storms outside midwinter, when rain falls on snow high up. And since 1993 the barriers have tended to arrive after the flow, not before it.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Heat is secondary here** versus **heat still needs planning.** The DMC's October–December outlook is wet with below-normal maximums, and El Niño's central-Chile signal is rain. But no outlook covers January–February yet, most of Santiago's heat records have fallen in El Niño years, and the Metropolitan Region was on red heat alert at 37 °C on 9 January 2026. The verifier downgraded "do not plan for heat" to a caution: put rain and slopes first through December, and keep the heat steps in action 6 ready for summer. A January–March DMC outlook would settle it.
- **Clearing settlements reduces risk** (the municipality's position, backed by a fire-service report) versus **clearing settlements moves risk** (Techo-Chile, and the skeptic, philanthropist, and frontline lenses). Knowing where the Las Zorras families now live would settle it; that is action 7.
- **Fund the river works** versus **fund the fire edge.** The historian and emergency manager put the nets and quebradas first; the funder argues a CLP 165 billion municipality should pay for river works, so private money should go where public money will not. Both are kept: public money for action 2, private money for actions 5 and 7.

* Suggested formulation: check with a health worker before sharing.

WHAT NOBODY COVERED

BLIND SPOT

Visitors. Lo Barnechea holds three ski areas (Farellones, La Parva, and Valle Nevado) and the road up to them, and the ski season runs into November. Thousands of people on a given day are neither residents nor on any register; they do not know the quebradas and they push past road closures. No lens researched how many are on the mountain on a storm day, or who is responsible for getting them down. Second gap: the comuna is too small to detect excess deaths. The open weekly death series is published by region only, and about 10–15 deaths a week in 132,000 people means a heat wave would be lost in the noise, so the region-level series run by SEREMI de Salud RM is the only way anyone will know whether summer heat killed here.

HOW TO READ THIS

METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people. They are not neighbourhood figures. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The request gave a street address in La Dehesa, Lo Barnechea, Región Metropolitana. It was widened to the comuna of Lo Barnechea, the smallest unit with data. Climate Impact Lab impact region, as the report records it: not resolved; the map has no searchable region list, and its downloadable data stops at country level. Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Heat is not the dominant hazard here, so the historical record carries the debris-flow, flood, water, and fire hazards instead of a borrowed number.

* Suggested formulation: check with a health worker before sharing.

- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources, mostly Chilean and in Spanish, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Six verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed. One verifier also settled conflicting figures for the record (the Las Condes damage count, the 2009 toll, and the 2016 water cut).
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not consulted this run: the inventory renders empty without its interface, and its download sits behind a registration form that was not filled in on anyone's behalf. No action here carries a checked precedent from it.
- **Health wording.** Passages where a wording error could hurt someone (when to leave ahead of a debris flow or a wildfire) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so. Every source is listed below.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. The impact region was not resolved. No published rainfall threshold for debris flows in the Santiago foothills was found. Who

* Suggested formulation: check with a health worker before sharing.

owns the drained fire-water pools on Cerro 18, where the cleared Las Zorras families now live, and who supplies water to Farellones could not be confirmed. Several local sources are press reports rather than official records.

References · Debris flows and alerts: Emol, Las Condes balance, 11 Oct 2026 · Emol, Senapred red alert, 10 Oct 2026 · 24horas, why no SAE alert · El Desconcierto, government on SAE · Radio Agricultura, Lo Barnechea monitoring and G-21 cuts · The Clinic, Lo Barnechea's nine nets · Emol, Las Condes barriers not installed · The Clinic, G-21 closure ignored · 2009 Lo Barnechea debris flow · Cooperativa, 6 Sep 2009 · BCN, Quebrada de Macul · Radio Agricultura, Macul basins 2022 · Emol, Jan 2021 debris flows · NHESS 2015, ring-net field tests · La Tercera, 1982 floods
Water: Meganoticias, El Huinganal cut 2021 · El Dínamo, store water before turbidity · El Dínamo, La Dehesa cut 2016 · T13, April 2016 cut · Infraestructura Pública, Pirque reserves
Fire: Urrutia-Jalabert et al. 2018 · Syphard et al. 2011 · T13, Chaguay accelerant · Pauta, Cerro 18 eviction and pools · 24horas, Cerro 18 fire 2022 · Forbes Chile, Viña toll 138 · IFRC operation update, Chile fires · The Clinic, Viña SAE complaints · GSMA, Lumkani evaluation
Settlements and people: The Clinic, Las Zorras final stage · The Clinic, La Cancha · Chilevisión, rural families 2021 · SERMIG, Lo Barnechea · Lo Barnechea older-adults programme 2026
Health and climate: UC / Lancet Countdown, extreme heat in Chile · Diario Oficial, heat-emergency annex · Hess et al. 2018, Ahmedabad · Meteored, DMC Oct–Dec 2026 outlook · T13, red heat alert Jan 2026 · datos.gob.cl, MINSAL weekly deaths
Food: UDP, June 2023 prices · ODEPA daily bulletin · ChileAtiende, PAE · CNN Chile, Escuelas Abiertas de Verano
Finance: Lo Barnechea budget report, June 2024 · ChileAtiende, Bono de Recuperación · Meganoticias, how FIBE works · Interior, Las Condes catastrophe zone · Hacienda, Ley 20.444 · World Bank, Chile cat bond 2023 · IFRC, Araucanía wildfire EAP · CMF, parametric insurance rule · Emol, Río Molina yellow alert · DMC, Mapocho en Los Almendros station · Hallegatte 2012 · Techo-Chile on Cerro 18
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory
Generated 15 October 2026 for Lo Barnechea, Santiago. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

* Suggested formulation: check with a health worker before sharing.